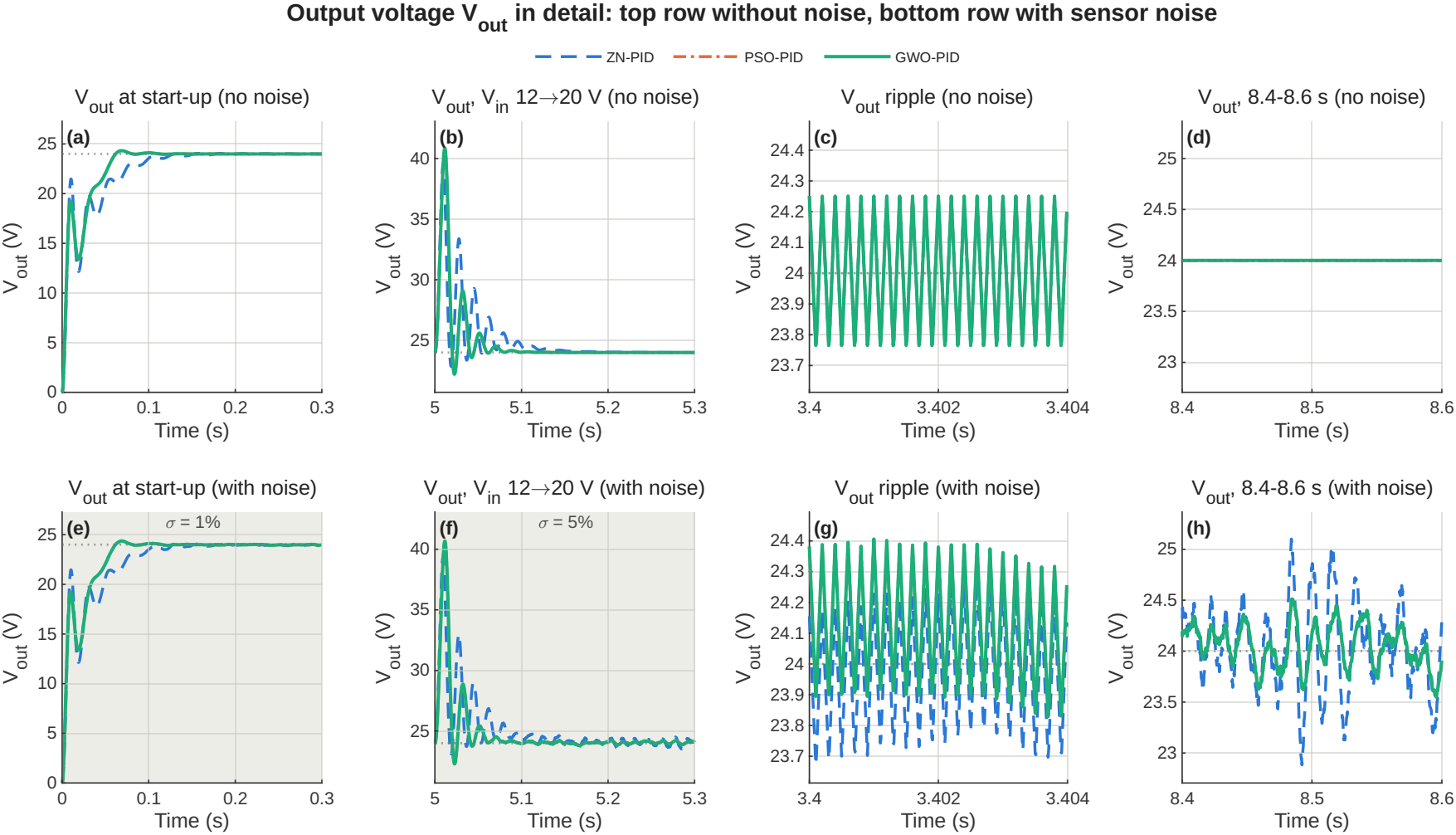

Output voltage V_out in detail: top row without noise, bottom row with sensor noise
ZN-PID
PSO-PID
GWO-PID
V_out at start-up (no noise)
V_out, V_in 12→20 V (no noise)
V_out ripple (no noise)
V_out, 8.4-8.6 s (no noise)
V_out at start-up (with noise)
V_out, V_in 12→20 V (with noise)
V_out ripple (with noise)
V_out, 8.4-8.6 s (with noise)
(a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)
σ = 1%
σ = 5%
V_out (V)
Time (s)